

Benqi Collateral Migrator Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

Immeas

April 6, 2025

Contents

1 About Cyfrin **2**

2 About This Document **2**

3 Disclaimer **2**

4 Copyright and License **2**

5 Audit Overview **3**

6 Audit Scope **3**

7 Risk Classification **3**

8 Protocol Summary **4**

8.1 Key Components 4

8.2 Collateral Migration Flow 4

9 Attestation **4**

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Benqi on Benqi Collateral Migrator. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated April 6, 2025, was delivered to Benqi, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from Benqi or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 4 days, the Cyfrin team conducted an audit on the Benqi Collateral Migrator code provided by Benqi. The findings consist of 4 Low severity issues with the remainder being informational.

Summary

Project Name	Benqi Collateral Migrator
Repository	benqi-collateral-migrator
Commit	d91ce3dbf56d...
Fix Commit	afe450e39083...
Audit Timeline	Mar 24th - Mar 27th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	4	4	0
Informational	2	2	0
Gas Optimizations	0	0	0
Total	6	6	0

6 Audit Scope

```
contracts/CollateralMigrator.sol
contracts/modules/SwapModule.sol
```

7 Risk Classification

	Impact: High	Impact: Medium	Impact: Low
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Benqi Collateral Migrator is a lightweight contract designed to help users of Benqi Markets seamlessly migrate the collateral of an existing position.

Ordinarily, a user would need to manually:

- Repay their loan,
- Withdraw their collateral,
- Swap it to the desired asset,
- Re-deposit it as collateral, and
- Reopen the loan.

The `CollateralMigrator` contract consolidates this entire process into a single transaction by leveraging flash loans from Trader Joe liquidity pairs and swaps via the 1inch DEX aggregator.

8.1 Key Components

- `CollateralMigrator`: The core contract responsible for executing collateral migrations. It enables users to swap the underlying collateral of their Benqi positions.

8.2 Collateral Migration Flow

1. The end user obtains swap data from the 1inch API.
2. The end user calls `CollateralMigrator::migrateCollateral` with the required swap parameters.
3. `CollateralMigrator` takes a flash loan from a Trader Joe liquidity pair.
4. Using the flash loan, `CollateralMigrator` opens a position in the target market.
5. With the position now overcollateralized, the original source market position is closed.
6. The source tokens are swapped for target tokens.
7. The target tokens are used to repay the flash loan along with the associated fees.
8. Any remaining target tokens are deposited into the target market on behalf of the user.

9 Attestation

Cyfrin attests that it performed a security review of the Benqi Collateral Migrator code provided by Benqi.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-04-06-cyfrin-benqi-collateral-migrator-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Benqi has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=16e63fd94b1b854a1bd4df07170fdc3833b221e574f6e99bb72e1482ca9be75b
F=2025-04-06-cyfrin-benqi-collateral-migrator-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.